Foundational Challenges in Dynamic Graph Representation Learning for Robust Network Telemetry Under Continuous Distribution Shifts

PhD Technical Research Proposal | Target Degree: Doctor of Philosophy (PhD) in Computer Science

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Core Areas: Graph Machine Learning, Self-Supervised Learning, Robust Network Telemetry, Causal Systems Provenance

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1. Problem Formulation: Reliability and Evaluation Failures in Security Data Science

Classifiers trained on static network intrusion benchmarks routinely report F1 scores above 0.99. Once deployed in live enterprise environments, these models degrade rapidly within weeks. This failure stems from four methodological problems:

1. **Temporal Snooping in Evaluation:** Most published benchmarks evaluate models using random k-fold cross-validation or unconstrained shuffling. Network traffic is sequential and causal. Randomly partitioning flow records across time lets future attack signatures, protocol states, and topology leak into historical training data [1, 2]. Test metrics look near-perfect, but models fail when tested under forward-chronological distribution shifts.
2. **Extreme Operational Label Scarcity:** Fewer than 0.1% of flows in live enterprise networks ever receive verified ground-truth labels from human analysts. Supervised models trained on small curated subsets overfit to known attack artifacts and fail when adversaries modify their cadences or tooling.
3. **Semantics Destruction in Graph Self-Supervised Learning:** Self-Supervised Learning (SSL) can extract representations from unlabeled traffic, but standard graph contrastive augmentations (edge dropping, feature masking) assume static citation networks. On network telemetry, arbitrary perturbations violate protocol state machines: dropping SYN packets produces impossible half-open connections, and artificial timestamp jitter breaks round-trip time constraints.
4. **The Statistical-Semantic Disconnect:** Modern Network Intrusion Detection Systems (NIDS) treat packets and flow records as statistical samples detached from the operating system processes that produced them. Routine administrative jobs (system updates, database backups) generate false positives that overwhelm security teams, while living-off-the-land attacks remain invisible in flow headers.

2. Core Research Questions and Methodological Challenges

This research investigates four foundational problems in dynamic graph representation learning for network telemetry:

- **RQ1: Leakage-Free Temporal Graph Evaluation and Drift Bounding.** Network communication forms an asynchronous dynamic multigraph $\mathcal{G}(t) = (\mathcal{V}(t), \mathcal{E}(t), \mathcal{X}_v(t), \mathcal{X}_e(t))$, where nodes $\mathcal{V}(t)$ represent communicating endpoints and edges $\mathcal{E}(t)$ represent directed communication flows carrying protocol semantics. Treating edges as independent samples violates physical causality:

$$\mathcal{I}\left(\mathcal{X}_e(t+\delta);\, \hat{\mathbf{y}}_t\right) = 0 \quad \text{for temporal embargo interval } \delta \geq \tau_{\text{session}} \tag{1}$$

  Core Question: What formal conditions and minimal temporal embargo intervals δ guarantee zero information leakage in streaming multigraph evaluation? How does out-of-distribution performance drop $\Delta F_1(t)$ scale with topological graph edit distance, diameter expansion, and protocol-state drift across multi-month deployments?
- **RQ2: Protocol-State Invariant Self-Supervised Representation Learning.** Given extreme label scarcity ($< 0.1\%$), representations must be learned directly from unlabeled streaming telemetry without human annotation, constrained by valid protocol state machines Σ_{protocol}:

$$\mathcal{L}_{\text{contrast}} = -\log \frac{\exp(\mathbf{q}\cdot\mathbf{k}^{+}/\tau)}{\exp(\mathbf{q}\cdot\mathbf{k}^{+}/\tau) + \sum_{j=1}^{K}\exp(\mathbf{q}\cdot\mathbf{k}_j^{-}/\tau)}, \quad \text{s.t. } \mathcal{T}(e) \in \Sigma_{\text{protocol}},\ |\Delta t_{\text{jitter}}| \leq \sigma_{\text{RTT}} \tag{2}$$

  Core Question: How can contrastive learning be constrained by protocol finite-state machines (TCP handshakes, TLS negotiation) and microsecond round-trip time bounds to preserve protocol semantics, prevent dimensional collapse in large negative queues ($K = 65,536$), and maintain stable representations over drifting traffic regimes?
- **RQ3: Decoupling Multi-Scale Dynamics (Progression vs. Volumetric Bursts).** In temporal graph models, sequence self-attention functions as a context-smoothing prior with near-uniform entropy ($\mathcal{H} \to \ln L$), diluting instantaneous volumetric bursts across historical baseline windows:

$$\mathcal{H}(\alpha_t) = -\sum_{i=1}^{L}\alpha_{t,i}\ln\alpha_{t,i} \approx \ln L \quad \Longrightarrow \quad \mathbf{h}_{\text{diluted}}(t) = \sum_{i=1}^{L}\alpha_{t,i}\mathbf{v}_{t-i} \tag{3}$$

  Core Question: What are the mathematical limits of temporal self-attention in non-stationary graphs, and how can dynamic graph models decouple multi-phase sequence progression from high-frequency velocity bursts without inflating false alarm rates?
- **RQ4: Cross-Layer Causal Attribution Across Network and Host Boundaries.** Statistical network telemetry and deterministic host execution graphs operate at different granularities, time scales, and abstractions. Tracing every host system call causes audit log explosion and degrades kernel throughput:

$$\text{Trigger(eBPF)} = \mathbb{I}\left[\alpha_t(u,v)\cdot\|\mathbf{x}_e(t)\|_2 > \gamma_{\text{kernel}}\right], \quad \text{LatencyBudget} \leq 5\,\mu\text{s} \tag{4}$$

  Core Question: How can dynamic graph attention weights selectively trigger lightweight kernel-level eBPF provenance tracing on anomalous socket state transitions, establishing causal paths from network sockets to host process trees while pruning $> 90\%$ of redundant causality edges within a $< 5\,\mu$s per-flow budget?

3. Investigation Framework and Methodological Trajectory

The proposed doctoral research spans four sequential phases:

Table 1: **Four-Phase Investigation Trajectory and Target Venues**

Phase	Core Methodological Focus	Target Venues & Primary Deliverables
Phase 1: Soundness	Formalize leakage-free forward partitioning ($\delta \geq \tau_{\text{session}}$); quantify empirical decay curves under continuous distribution shift vs. cross-validation inflation.	**USENIX Security / IEEE S&P** Longitudinal benchmark suite & temporal splits
Phase 2: Representation	Protocol-State Invariant Augmentations (PSIA) respecting TCP state machines; curvature-aware graph rewiring to resolve multi-hop over-squashing.	**NeurIPS / ICLR** PyTorch Geometric stability toolkit
Phase 3: Systems	Selective eBPF provenance tracing triggered by dynamic GNN attention; causal path reduction isolating process trees under line-rate $< 5\,\mu$s latency.	**ACM CCS / NDSS** Functional eBPF-GNN defense engine
Phase 4: Synthesis	Enterprise campus red-teaming, distribution-shift benchmarking, and generalizable principle validation; complete doctoral dissertation authoring and defense.	**PhD Dissertation Defense** Containerized reproducibility package

Note on Mentorship and Project Scope: This trajectory provides a structured starting point. Doctoral research requires adapting hypotheses and priorities as empirical results develop, and I expect to refine these directions with my advisor.

4. Preliminary Empirical Observations Motivating the Research Questions

Preliminary experiments from my master's thesis expose these methodological problems on real traffic traces:

1. **Sequence Permutation Collapse (RQ1, RQ3):** In controlled sequence ablation tests on multi-stage attack progressions, shuffling sequence order ($\pi(t)$) caused a **27.57 percentage point drop** in multi-class Macro-F1 (0.7150 $\rightarrow$ 0.4393, a 38.6% relative collapse). Temporal attention entropy stayed flat ($\Delta\mathcal{H} = -0.0017$). This shows that sequential recurrent transitions, rather than permutation-invariant self-attention, capture multi-step attack progression.
2. **Volumetric Low-Pass Dilution (RQ3):** Standard temporal attention functions as a low-pass filter over high-volume traffic bursts, spreading burst signals across historical windows and producing a 30.54% false-negative rate. Adding velocity and snapshot skip connections lowered breach misses to **0.36%** (558/560 attacks stopped; Wilson 95% CI: [0.04%, 1.28%]).
3. **False Alarm Calibration (RQ4):** On enterprise NetFlow data, setting decision thresholds at the validation ROC inflection point ($\tau = 0.494$) reduced false alarms from 12.04% to **0.76%** (a 93.7% reduction) without missed breaches ($n = 3,517$). Eliminating the remaining false positives requires linking flow statistics directly to host process provenance.

5. Background Preparation, Computational Feasibility, and Risk Mitigation

Background Preparation: I developed ~12,000 lines of PyTorch Geometric and multi-threaded flow parsing code during my master's thesis; published three papers in Springer *Lecture Notes in Networks and Systems* (LNNS, 2025); implemented LLM quantization and vector search at the Financial Reporting Council (Ministry of Finance, Bangladesh); and hold an Amateur Radio license (**Callsign: S21NEO**) with practical experience in RF systems and telemetry.

Table 2: **Identified Methodological Risks and Concrete Scientific Mitigations**

Risk Area	Potential Failure Mechanism	Concrete Scientific Mitigation
Scale Explosion	Temporal training on millions of streaming edges per hour exhausts GPU memory.	Temporal random walk neighborhood sampling and gradient checkpointing to maintain $O(\lvert\mathcal{E}\rvert)$ linear complexity.
Semantic Drift	Unconstrained graph perturbations produce invalid packet sequences (such as orphaned handshakes).	Enforce Protocol-State Invariant Augmentations (PSIA) with sequence-number constraints and protocol state validation.
Kernel Overhead	Continuous system-call logging degrades gateway throughput on high-bandwidth links.	Selective provenance tracing triggered on anomalous socket transitions, bounding latency to $< 5\,\mu$s per flow.
Model Forgetting	Continual fine-tuning under drift degrades historical baseline representations.	Elastic weight consolidation (EWC) penalties on key attention layers paired with prototypical reservoir replay buffers.

References

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